INCH-POUND
A-A-50740A
8 July 2009
SUPERSEDING
A-A-50740
30 October 1987

COMMERICIAL ITEM DESCRIPTION
SPECTRUM ANALYZER
(10 kHz – 50 GHz)

Reinstated after (8 July 2009) and may be used for new and existing designs and acquisitions.

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This Commercial Item Description(CID) describes a solid-state, internal pre-selection, programmable, 10 Hz resolution bandwidth, 10 kHz to 50 GHz, Spectrum Analyzer including external mixers. This CID is meant as a minimum requirement for the Spectrum Analyzer in which only those manufacturers that meet or surpass the following requirements are qualified per this CID. Test equipment offered may be commercially available and may be modified to the extent necessary to meet the following requirements are supplied per this CID.

2. CLASSIFICATION. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800F, except as specified herein.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

3.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800F for the classification as stated herein except as specified herein.

3.2 Temperature Requirements. The temperature ranges for both operating and non- operating conditions shall be as specified for Class 3 equipment in MIL-PRF-28800F.

3.3 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 110/220 Volts AC (+10%) at line frequencies of 45 Hz to 440 Hz. Power consumption shall not exceed 650 VA. Transient state conditions shall meet MIL-PRF-28800F requirements for Class III equipment.

3.4 Magnetic Environment. The equipment shall comply with the magnetic environment requirements of MIL-PRF-28800F.

3.5 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 8000 hours of operation with a statistical certainty of 95%.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data that may improve this document should be sent to: WR-ALC/742 CBSG, 460 Richard Ray BLVD, Ste 200, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

FSC 6625

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3.6 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the equipment case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the equipment.

3.7 Performance Requirements. The equipment shall meet all requirements specified herein, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800F for the class specified herein.

3.7.1 Frequency Range. The minimum frequency range of the Spectrum Analyzer with internal and external mixers shall be 10 kHz to 50 GHz.

3.7.2 Frequency Reference Accuracy.

a. Aging rate. After 24 hours warm-up period, the aging shall be less than $1x10^{-9}$/day, and less than $2.5x10^{-7}$/year.

b. Temperature Stability. The temperature stability shall be better than $2x10^{-8}$ for any temperature change between 0° C and $+55^{o}$ C.

3.7.3 Center Frequency Accuracy. The center frequency accuracy at 1GHz with 0 span shall be better than +/-50 Hz/day; span/div. controls shall not effect accuracy more than +/-2% for spans less than 1 MHz.

3.7.4 Frequency Counter. The Spectrum Analyzer shall provide a fully integrated frequency counter capability. The frequency counter accuracy shall allow a 2 GHz signal to be measured within 400 Hz. The total accuracy shall be defined as: +[Counter Frequency X Reference Frequency Accuracy + (50 Hz X Harmonic Mixing Number) + 1 LSD].

3.7.5 Frequency Span. The minimum span range shall be 100 Hz to a frequency range of 21 GHz over 10 divisions of the display's horizontal axis. The frequency span accuracy shall be better than +/- 10% over all spans.

3.7.6 Resolution Bandwidth. The resolution bandwidth shall be 10 Hz to 3 MHz (3 or 6 dB). The setting shall be front-panel selectable or automatic.

3.7.7 Residual FM. The total residual FM shall be less than 10 Hz peak to peak in 20 milliseconds, using a 10 Hz resolution bandwidth in internal mixing for span setting less than or equal to 100 kHz in the phase lock mode.

3.7.8 Frequency Drift. The frequency drift after a 1-hour warm up period shall be less than 50 Hz/min. for frequency span less than 100 kHz.

3.7.9 Sensitivity. The following shall be a minimum acceptable overall sensitivity at 10 Hz resolution bandwidth and lowest harmonic-mixing mode: (Clause does not refer to individual frequency bands.)

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3.7.10 Sensitivity with Internal Mixer.

Frequency Range	Sensitivity (dBm)
10 kHz - 50 kHz	-95
50 kHz - 1.8 GHz	-110
1.8 GHz - 6.4 GHz	-122
6.4 GHz - 21 GHz	-102

3.7.11 Sensitivity with External Mixer.

Frequency Range	Sensitivity (dBm)
18 GHz - 26.5 GHz	-110
26.5 GHz - 40 GHz	-108
33 GHz - 50 GHz	-102

3.7.12 Frequency Response. When tuned to the specified frequency range, with attenuator set at 10 dB, the analyzer shall have the following:

3.7.13 Frequency Response with Internal Mixer.

Frequency Range	Response dB max
10 kHz - 1.8 GHz	+/- 1.5
1.8 GHz - 6.4 GHz	+/- 2.5
6.4 GHz - 13 GHz	+/- 3.5
13 GHz - 18 GHz	+/- 4.0
18 GHz - 21 GHz	+/- 5.0

3.7.14 Frequency Response with External Mixer.

Frequency Range	Response dB max
21 GHz - 26.5 GHz	+/- 2.5
26.5 GHz - 50 GHz	+/- 2.5

3.7.15 Sideband Noise. The Sideband noise shall be -86 dBc/Hz or less at 10 kHz offset from center frequency and in the lowest harmonic mixing mode.

3.7.16 Single Input Range. The minimum amplitude reference level range shall be from -115 dBm to +30 dBm with a safe input of +30 dBm in the log mode or 1 Watt in the linear mode.

3.7.17 Input Attenuator Accuracy. The input Attenuator shall have a minimum range of 0-60 dB with front panel selectable steps of 10 dB attenuator. The attenuator accuracy shall not exceed 3.0 dB over 60 dB range from 0-18 GHz, and 6.0 dB over the 60 dB range overall from 18-21 GHz.

3.7.18 RF Input Connector. For frequencies up to 21 GHz, the RF input connector shall be compatible with the Type N connectors. When mixers are used, the connection shall be through a combination of waveguide mixers, adapters and connectors. The RF input connectors shall be compatible and mate nondestructively with WR 42 Flange UG-595/U (18-26.5 GHz), WR 28 Flange UG-500/U (26.5-40 GHz), and WR 22 Flange UG-383/U (33-50 GHz).

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3.7.19 Input Impedance. The input impedance shall be 50 ohms nominal. The Voltage Standing Wave Ratio (VSWR) shall not be greater than 3:1 up to 18 GHz; nor greater than 3.5:1 from 18 GHz to 21 GHz.

3.7.20 Input Power. The Spectrum Analyzer shall accept alternating signals that are less than 1 watt (+30 dBm) continuous over the frequency range of 10 kHz - 21 GHz. The waveguide input over the frequency range of 18 - 50 GHz shall accept alternating signals that are less than or equal 100 mW (+20 dBm) continuous.

3.7.21 External Reference Input. The equipment shall be provided with an input to accommodate a 10 MHz external reference.

3.7.22 Output. The equipment shall be provided with an internal calibrator signal output, an X-Y recorder output, 1st Local Oscillator output greater than +5 dBm and 50 ohm nominal impedance; a pen-lift control, and an IF output at –5 dBm level and 50 ohm nominal impedance.

3.7.23 Vertical Scaling. Linear or log vertical scaling shall be selectable from the front panel. The log scale shall have a minimum of 2 selectable calibrated scales: 2dB/div and 10db/div.

3.7.24 Display Accuracy. The maximum allowable cumulative error in the log display mode shall be +/- 4.0 dB. The maximum accuracy at full scale display, in the linear mode shall be +/- 5%.

3.7.25 Third Order Intermodulation Distortion. The third order intermodulation distortion shall be -70 dBc or less at 5 GHz with an input level to the mixer of -30 dBm for each tone.

3.7.26 Harmonic Distortion. The harmonic distortion to 2 GHz shall be -60 dBc or less, and -100 dBc or less from 2 GHz to 21 GHz with an input level to the mixer of -40 dBm.

3.7.27 Local Oscillator Emissions. The Local Oscillator emissions referenced to the input mixer shall be -80 dBm or lower with 10 dB RF attenuation.

3.7.28 Digital Storage. Digital storage shall be provided with selectable modes that save maximum signal values and noise-average spectral displays of two signals. The digital storage function shall be capable of storing and displaying at least eight spectrums including LCD readout measurement parameters. When a battery is required for the digital storage circuitry, the battery life shall be at least 12 months under the normal operating conditions.

3.7.29 Markers. The equipment shall provide frequency and amplitude markers, using one marker for absolute measurements and two markers for relative measurements. Marker positioning shall provide readout of frequency and amplitude of any point, or the difference in amplitude or frequency between two points within the frequency range of the instrument.

3.7.30 Display Viewing Area. The equipment shall provide a flat display, with a minimum useful scan area of 10 divisions wide by 8 divisions high. The graticule will include a log scale of 10, 5, 2, and 1 dB/div and a linear scale of 10% of reference level/div.

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3.7.31 Display Readout. The Display shall be capable of displaying the following readout:

a. Reference level.

b. Center frequency.

c. Vertical display mode.

d. Frequency span/div. or frequency span.

e. Resolution bandwidth.

f. RF attenuation.

3.7.32 Mixers. The Spectrum Analyzer shall use no more than three (3) mixers and, if required, a unit driving amplifier. The ranges and conversion loss shall be as follows:

Mixer Frequency Range	Average Conversion Loss Maximum
18 – 26.5 GHz	-30 dB
26.5 – 40 GHz	-30 dB
33 – 50 GHz	-30 dB

3.7.33 Interface. The Spectrum Analyzer shall provide an IEEE standard 488, USB, and GPIB interface function and connector. This is to enable the Spectrum Analyzer for remote control of all front panel function except power switch, display intensity, frequency calibration, level calibration and trigger level adjustment knob.

3.7.34 Programmability. The Spectrum Analyzer shall be fully programmable except for display positioning, display visual controls, manual scan, level calibration and power.

3.7.35 Front-Panel Controls. The following shall be front-panel controllable:

a. Store display
b. Erase display
c. Operate in non-storage display mode
d. Store and display two traces, one active and one previously stored spectrum
e. Perform normalization or trace subtraction
f. Peak hold
h. Selective display and blanking
i. Internal preselection
j. Digital averaging
k. Vertical display
l. Linear display
m. Logarithmic display
n. Intensity

3.7.36 Signal Identification. The instrument shall have the capability, when using external mixers, of identifying a true signal from a false signal when preselection is not available. This capability may be selected from a front panel control or from menu selection.

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3.7.37 Accessories. Each instrument shall be provided with the following:

a. Spare fuses
b. 115V power cable
c. Operator manual
d. Programmer manual
e. 18 - 26.5 GHz mixer (if needed to meet specifications)
f. 26.5 - 40 GHz mixer
g. 33 - 50 GHz mixer
h. Mixer amplifier (if required)
i. N male to BNC female adapter
j. BNC to BNC connector, 50 ohm coax cable
k. 2 meter, double shielded GPIB cable

3.7.38 Dimensions and Weight. The size and weight of the Spectrum Analyzer shall be consistent with current commercial practices.

4. REGULATORY REQUIREMENTS.

4.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

6. PACKAGING.

6.1 Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Source of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

7.1.2 The Federal Acquisition Regulation (FAR) may be obtained at www.acqnet.gov/far or from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

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7.1.3 Institute of Electrical and Electronics Engineers (IEEE) standards products may be obtained at http://www.ieee.org or available from IEEE Operations Center. Phone: (800)678-IEEE. Mail: IEEE Operations Center, Sales Office, 445 Hoes Lane, PO Box 1331, Piscataway, NJ, 08855-1331, USA.

7.1.4 ANSI/EIA standards can be obtained at http://www.ansi.org or http://www.eia.org or available from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

7.2 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Product conformance provisions.
c. Packaging requirements.

7.3 Key words.

Output
Signal source
Testing purposes

MILITARY INTERESTS:

Custodians:
Army – MI
Navy – SH
Air Force – 84

Review activities:
Army – GL
Air Force – 99
DLA – CC

Preparing activity:
Air Force – 84

Agent:
Air Force – 99

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